AI & COMPUTING

Curiosa mathematica, part II : Pillow-problems thought out during wakeful hours

Lewis Carroll

1895

PILLOW-PROBLEMS

[See p. 100.

Curiosa Mathematica

PART II

PILLOW-PROBLEMS

THOUGHT OUT DURING

WAKEFUL HOURS

BY

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FOURTH EDITION

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[Pg vii]

PREFACE TO FOURTH EDITION.

I TAKE this opportunity of explaining why it is that (as stated in the Note to p. xix) I have used the symbols and to represent the words '' and ''.

The use of some symbols needs, I suppose, no more justification than the use of + and - to represent '' and ''.

These particular symbols are derived from the old theory of Trigonometry, in which sines, cosines, &c. were actual lines.

In this diagram, being taken as the unit of length, is the sine of the angle , and its cosine.

In each of my two symbols I have retained the semicircle in the symbol , I have merely moved to the middle; and, in the symbol , I have lengthened , taking it a little beyond the curve, in order to avoid confusion with the existing symbol for ''.

I also take this opportunity of adding a sort of Corollary (lately thought out) to the solution of Problem 59 (see p. 84).

[Pg viii]

If , , be given lengths, they must, in order that the Tetrahedron may be possible, fulfil certain conditions, as follows:—

(1) they have to form the sides of a Triangle: hence any two of them must be greater than the third;

(2) the three angles of this Triangle have to form a solid angle: hence any two of these angles must be together greater than the third: hence any two of them must be together greater than 90°: hence any one of them must be less than 90°: hence the cosine of any one of them must be greater than 0: i.e. must be greater than O, &c.: hence , , must be such that the squares of any two of them are together greater than the square of the third.

For example, the lengths 2, 3, 4 would not do as the given lengths, since, although fulfilling the first condition, by having , they fail to fulfil the second, as is not .

C. L. D.

CH. CH., OXFORD.

March, 1895.

[Pg ix]

PREFACE TO SECOND EDITION.

THE principal changes, made in this Second Edition of "Pillow-Problems", are as follows:—

(1) After the numeral, which precedes each Question, Answer, or Solution, references are given to the pages at which the corresponding matter may be found.

(2) Some of the Solutions have been re-arranged, and duplicate-diagrams have been inserted, in order that every portion of text may have its illustrative diagram visible along with it, and that the reader may thus be saved the trouble, and the strain on his temper, involved in turning a leaf backwards and forwards while referring from the one to the other.

(3) In the title of the book, the words "sleepless nights" have been replaced by "wakeful hours".

This last change has been made in order to allay the anxiety of kind friends, who have written to me to express their sympathy in my broken-down state of health, believing that I am a sufferer from chronic "insomnia", and that it is as a remedy for that exhausting malady that I have recommended mathematical calculation.

The title was not, I fear, wisely chosen; and it certainly was liable to suggest a meaning I did not intend to convey,[Pg x] viz. that my "nights" are very often wholly "sleepless". This is by no means the case: I have never suffered from "insomnia": and the over-wakeful hours, that I have had to spend at night, have often been simply the result of the over-sleepy hours I have spent during the preceding evening! Nor is it as a remedy for wakefulness that I have suggested mathematical calculation; but as a remedy for the harassing thoughts that are apt to invade a wholly-unoccupied mind. I hope the new title will express my meaning more lucidly.

To state the matter logically, the dilemma which my friends suppose me to be in has, for its two horns, the endurance of a sleepless night, and the adoption of some recipe for inducing sleep. Now, so far as my experience goes, no such recipe has any effect, unless when you are sleepy: and mathematical calculation would be more likely to delay, than to hasten, the advent of sleep.

The real dilemma, which I have had to face, is this: given that the brain is in so wakeful a condition that, do what I will, I am certain to remain awake for the next hour or so, I must choose between two courses, viz. either to submit to the fruitless self-torture of going through some worrying topic, over and over again, or else to dictate to myself some topic sufficiently absorbing to keep the worry at bay. A mathematical problem is, for me, such a topic; and is a benefit, even if it lengthens the wakeful period a little. I believe that an hour of calculation is much better for me than half-an-hour of worry.

The reader will, I think, be interested to see a curiously illogical solution which has been proposed, by a correspondent[Pg xi] of the Educational Times, for Problem 61, viz. "Prove that, if any 3 Numbers be taken, which cannot be arranged in , and whose sum is a multiple of 3, the sum of their squares is also the sum of another set of 3 squares, the 2 sets having no common term."

The proposed solution is as follows:—

"Let , , be the three Numbers; then

Also

"

Now, if we denote, by '', the property "which cannot be arranged in , and whose sum is a multiple of 3," and, by '', the property "the sum of whose squares is also the sum of another set of 3 squares, the 2 sets having no common term," we see that all, that this writer has succeeded in proving, is that certain selected Numbers, which have property '', have also property '': but this does not prove my Theorem, viz. that any Numbers whatever, which have property '', have also property ''. If his argument were arranged in a syllogistic form, it would be found to assume a quite untenable Major Premiss, viz. "that, which is true of certain selected Numbers which have

property '', is true of any Numbers whatever which have property ''."

C. L. D.

CH. CH., OXFORD.

September, 1893.

[Pg xii]

INTRODUCTION.

NEARLY all of the following seventy-two Problems are veritable "Pillow-Problems", having been solved, in the head, while lying awake at night. (I have put on record the exact dates of some.) No. 37 and one or two others belong to the daylight, having been solved while taking a solitary walk; but every one of them was worked out, to the very end, before drawing any diagram or writing down a single word of the solution. I generally wrote down the answer, first of all: and afterwards the question and its solution. For example, in No. 70, the very first words I wrote down were as follows:—"(1) down back-edge; up again; down again; and so on; (2) about of the way down the back-edge; (3) about 18° 18′; (4) about 14°." These answers are not quite correct; but at least they are genuine, as the results of mental work only. "A poor thing, Sir, but mine own!"

My motive, for publishing these Problems, with their mentally-worked solutions, is most certainly not any desire to display powers of mental calculation. Mine, I feel sure, are nothing out-of-the-way; and I have no doubt there are many mathematicians who could produce, mentally, much[Pg xiii] shorter and better solutions. It is not for such persons that I intend my little book; but rather for the much larger class of ordinary mathematicians, who perhaps have never tried this resource, when mental occupation was needed, and who will, I hope, feel encouraged—by seeing what can be done, after a little practice, by one of average mathematical powers—to try the experiment for themselves, and find in it as much advantage and comfort as I have done.

The word "comfort" may perhaps sound out of place, in connection with so entirely intellectual an occupation; but it will, I think, come home to many who have known what it is to be haunted by some worrying subject of thought, which no effort of will is able to banish. Again and again I have said to myself, on lying down at night, after a day embittered by some vexatious matter, "I will not think of it any more! I have gone through it all, thoroughly. It can do no good whatever to go through it again. I will think of something else!" And in another ten minutes I have found myself, once more, in the very thick of the miserable business, and torturing myself, to no purpose, with all the old troubles.

Now it is not possible—this, I think, all psychologists will admit—by any effort of volition, to carry out the resolution "I will not think of so-and-so." (Witness the common trick, played on a child, of saying "I'll give you a penny, if you'll stand in that corner for five minutes, and not once think of strawberry-jam." No human child ever yet won the tempting wager!) But it is possible—as I am most thankful to know—to

carry out the resolution "I will[Pg xiv] think of so-and-so." Once fasten the attention upon a subject so chosen, and you will find that the worrying subject, which you desire to banish, is practically annulled. It may recur, from time to time—just looking in at the door, so to speak; but it will find itself so coldly received, and will get so little attention paid to it, that it will, after a while, cease to be any worry at all.

Perhaps I may venture, for a moment, to use a more serious tone, and to point out that there are mental troubles, much worse than mere worry, for which an absorbing subject of thought may serve as a remedy. There are sceptical thoughts, which seem for the moment to uproot the firmest faith; there are blasphemous thoughts, which dart unbidden into the most reverent souls; there are unholy thoughts, which torture, with their hateful presence, the fancy that would fain be pure. Against all these some real mental work is a most helpful ally. That "unclean spirit" of the parable, who brought back with him seven others more wicked than himself, only did so because he found the chamber "swept and garnished", and its owner sitting with folded hands: had he found it all alive with the "busy hum" of active work, there would have been scant welcome for him and his seven!

My purpose—of giving this encouragement to others—would not be so well fulfilled had I allowed myself, in writing out my solutions, to improve on the work done in my head. I felt it to be much more important to set down what had actually been done in the head, than to supply shorter or neater solutions, which perhaps would be much harder to do without paper. For example, a Long-Multiplication[Pg xv] sum (say the multiplying together of two numbers of 7 digits) is no doubt best done, on paper, by beginning at the unit-end, and writing out 7 rows of figures, and adding up the columns in the usual way. But it would be very difficult indeed—to me quite impossible—to do such a thing in the head. The only chance seems to be to begin with the millions, and get them properly grouped; then the hundred-thousands, adding the results to the previous one; and so on. Very often it seems to happen, that the easiest mental process looks decidedly lengthy and round-about when committed to paper.

When I first tried this plan, easy geometrical problems were all I could manage; and, even in these, I had to pause from time to time, in order to re-draw the diagram, which would persist in getting 'rubbed-out'. Algebraical problems I avoided at first, owing to the provoking fact that, if one single co-efficient escaped the memory, there was no resource but to begin the calculation all over again. But I soon got over both these difficulties, and found myself able to remember fairly large numerical co-efficients, and also to retain, in the mind's eye, fairly complex diagrams, even to the extent of finding my way from one part of the diagram to another. The lettering of the diagrams proved such a troublesome thing to keep in the memory, that I almost gave up using it, and learned to recognise Points by their situation only. In my MS. of No. 53, I find the following memorandum:—

"I had never set myself this Problem before the week ending Ap. 6, 1889. I tried it, two or three nights, lying awake; and finally worked it out on the night of Ap. .[Pg xvi] All the conclusions were worked out mentally before any use was made of pen and paper. While working it, I did not give names to any Points, except , , , and : I merely thought of them by their positions (e. g. 'the foot of the perpendicular from on ')."

If any of my readers should feel inclined to reproach me with having worked too uniformly in the region of Common-place, and with never having ventured to wander out of the beaten tracks, I can proudly point to my one Problem in 'Transcendental Probabilities'—a subject in which, I believe, very little has yet been done by even the most enterprising of mathematical explorers. To the casual reader it may seem abnormal, and even paradoxical; but I would have such a reader ask himself, candidly, the question "Is not Life itself a Paradox?"

To give the Reader some idea of the process of construction of these Problems, I will give the biography of No. 63. The history of one is, to a great extent, the history of all.

It was begun during the night of Sept. , 1890, and completed during the following night. The idea had occurred to me, a short time previously, that something interesting might be found in the subject of what I may call 'partially-regular' Solids. The 'regular' Solids are provokingly few in number; and it would be hopeless to find any question, connected with them, that has not already been exhaustively analysed: some also of the 'partially-regular' Solids (e. g. rhomboidal crystals) have probably been similarly treated; but there seemed to be room for the invention of other such Solids.

[Pg xvii]

Accordingly, I devised a Solid enclosed, above and below, by 2 equal and parallel Squares, having their centres in the same vertical line, and the upper one twisted round so that its sides should be parallel to the diagonals of the lower Square. Then I imagined the upper one raised until its corners formed the vertices of 4 equilateral Triangles, whose bases were the sides of the lower one. The Solid, thus obtained, was evidently enclosed by 2 Squares and 8 equilateral Triangles: and the Problem I set myself was to obtain its Volume.

There was no great difficulty in proving that the distance between the 2 Squares (taking each side as equal to '2') was . But, when I looked about for some Trigonometrical method for calculating the Volume, despair soon seized upon me! A calculable Prism could be cut out of the middle of the Solid, I saw: but the outlying projections completely baffled me. After a while, the happy idea occurred to me of trying Algebraical Geometry, and regarding each facet as the base of a Pyramid, having its vertex at the centre of the Solid, which I decided to take as the Origin. I saw at once that I could calculate the co-ordinates of all the vertical Points, thence obtain equations to the Planes containing the facets, and thence calculate their distances from the Origin, which would be the altitudes of the Pyramids. Also it was evident that a sample Pyramid would suffice. I worked out a value for the Volume, that first night; but the thing got into a tangle, and I felt pretty sure I had got it wrong.

The next night I began again, and worked it all through from the beginning. In the morning the answer[Pg xviii] was clear in my memory, and I wrote it down at once; and did not write out the Problem, and its solution, until later in the day, when I was well pleased to find the written proof confirm the result I had arrived at in the hours

of darkness.

It is not, perhaps, much to be wondered at that, when these Problems came to be re-written and arranged for publication, a good many mistakes were discovered. Some were so bad as quite to spoil the solutions in which they occurred: these Problems I have omitted altogether. The others I have corrected, in the solutions as given in Chapter III: but, that I may not be credited with an amount of accuracy, as a computator, which I am well aware I do not possess, I here append a list of them.

In No. 7, in the denominator '', I forgot the '2'.[1]

In No. 10, I failed to notice that the 3 coins might also be a half-crown and 2 shillings.

In No. 13, in the last line but one, I put '', instead of ''.

In No. 32, I brought out the arithmetical value as '358520', instead of '358550'.

In No. 38, I got the decimal wrong, making it instead of , and thus brought out the answer as instead of .

In No. 44, I said that the denominator would be of the form . This last factor is superfluous: i. e. .

[Pg xix]

In No. 50, I made a mistake near the end, bringing out , instead of .

In No. 55, I put 'tan' for 'sin'.

In No. 57, in the last paragraph, I replaced the denominator '' by (what I imagined to be its equivalent) ''. Apparently I was under the delusion that '' was the same thing as ''!

In No. 70, section (3), I forgot to add in the , thus making the answer half a degree wrong. And, in section (4), I forgot to add in the , thus again making the answer half a degree wrong.

Let me, in conclusion, gratefully acknowledge the valuable assistance I have received from Mr. F. G. Brabant, M.A., of Corpus Christi College, Oxford, who has most patiently and carefully gone through my proofs, first working out each result independently, and has thus detected many mistakes which had escaped my notice. He has also supplied, for No. 59, a much neater answer than mine, viz. .

Other mistakes may perchance, having eluded us both, await the penetrating glance of some critical reader, to whom the joy of discovery, and the intellectual superiority which he will thus discern, in himself, to the author of this little book, will, I hope, repay to some extent the time and trouble its perusal may have cost him!

C. L. D.

CH. CH., OXFORD.

May, 1893.

FOOTNOTES:

[Pg xx]

[1] In the trigonometrical Problems, I have used the symbols and , to represent the words 'sine' and 'cosine'.

CONTENTS.

PAGE

CHAPTER I. Questions 1

CHAPTER II. Answers 19

CHAPTER III. Solutions 28

[Pg xxi]

SUBJECTS CLASSIFIED.

Arithmetic. No. 31.

Algebra:—

Equational Problems. Nos. 8, 25, 39, 52, 68.

Series. Nos. 21, 32.

Indeterminate Equations. No. 47.

Properties of Numbers. Nos. 1, 14, 29, 44, 61.

Chances. Nos. 5, 10, 16, 19, 23, 27, 38, 41, 45, 50, 58, 66.

Pure Geometry, Plane. Nos. 2, 3, 9, 15, 17, 18, 20, 24, 26, 30, 34, 35,

36, 40, 46, 51, 57, 62, 64, 71.

Trigonometry:—

Plane. Nos. 4, 6, 7, 11, 12, 13, 18, 22, 28, 37, 42, 43, 48, 54, 55, 56,

57, 60, 65, 69.

Solid. Nos. 49, 59, 63, 70.

Algebraical Geometry:—

Plane. No. 53.

Solid. No. 67.

Differential Calculus:—

Maxima and Minima. No. 33.

Transcendental Probabilities. No. 72.

PILLOW-PROBLEMS.

[Pg 1]

CHAPTER I.

Questions.

1. (28)[2]

Find a general formula for two squares whose sum = 2.

[24/3/84

2. (29)

In a given Triangle to place a line parallel to the base, such that the portions of sides, intercepted between it and the base, shall be together equal to the base.

3. (30)

If the sides of a Tetragon pass through the vertices of a Parallelogram, and if three of them are bisected at those vertices: prove that the fourth is so also.

4. (30)

In a given acute-angled Triangle inscribe a Triangle, whose sides make, at each of the vertices, equal angles with the sides of the given Triangle.

[19/4/76

[Pg 2]

5. (19, 31)

A bag contains one counter, known to be either white or black. A white counter is put in, the bag shaken, and a counter drawn out, which proves to be white. What is now the chance of drawing a white counter?

[8/9/87

6. (19, 32)

Given lengths of lines drawn, from vertices of Triangle, to middle points of opposite sides, to find its sides and angles.

7. (19, 33)

Given 2 adjacent sides, and the included angle, of a Tetragon; and that the angles, at the other ends of these 2 sides, are right: find (1) remaining sides, (2) area.

[4 or 5/89

8. (20, 34)

Some men sat in a circle, so that each had 2 neighbours; and each had a certain number of shillings. The first had 1/ more than the second, who had 1/ more than the third, and so on. The first gave 1/ to the second, who gave 2/ to the third, and so on, each giving 1/ more than he received, as long as possible. There were then 2

neighbours, one of whom had 4 times as much as the other. How many men were there? And how much had the poorest man at first?

[3/89

9. (35)

Given two Lines meeting at a Point, and given a Point lying within the angle contained by them: draw, from the given Point, two lines, at right angles to each other, and[Pg 3] forming with the given Lines and the line joining their intersection to the given Point, two equal Triangles.

[11/76

10. (20, 36)

A triangular billiard-table has 3 pockets, one in each corner, one of which will hold only one ball, while each of the others will hold two. There are 3 balls on the table, each containing a single coin. The table is tilted up, so that the balls run into one corner, it is not known which. The 'expectation', as to the contents of the pocket, is . What are the coins?

[8/90

11. (20, 36)

A Triangle has another inscribed in it, so that ; thus making it similar to the first Triangle. Find ratio between homologous sides. And solve for "".

The Triangles can be proved similar thus:—

Hence

Let . We have to find .

[31/3/82

[Pg 4]

12. (20, 37)

Given the semi-perimeter and the area of a Triangle, and also the volume of the cuboid whose edges are equal to the sides of the Triangle: find the sum of the squares of its sides.

[23/1/91

13. (20, 38)

Given the lengths of the radii of two intersecting Circles, and the distance between their centres: find the area of the Tetragon formed by the tangents at the points of intersection.

[3/89

14. (39)

Prove that 3 times the sum of 3 squares is also the sum of 4 squares.

[2/12/81

15. (39)

If a Figure be such that the opposite angles of every inscribed Tetragon are supplementary: the Figure is a Circle.

[3/91

16. (20, 40)

There are two bags, one containing a counter, known to be either white or black; the other containing 1 white and 2 black. A white is put into the first, the bag shaken, and a counter drawn out, which proves to be white. Which course will now give the best chance of drawing a white—to draw from one of the two bags without knowing which it is, or to empty one bag into the other and then draw?

[10/87

17. (40)

In a given Triangle place a line parallel to the base, such that if, from its ends, lines be drawn, parallel to the[Pg 5] sides and terminated by the base, they shall be together equal to the first line.

[3/89

18. (21, 41)

Find a Point, in the base of a given Triangle, such that, if from it perpendiculars be dropped upon the sides, the line joining their extremities shall be parallel to the base. (1) Trigonometrically. (2) Geometrically.

[11/89

19. (21, 42)

There are 3 bags; one containing a white counter and a black one, another two white and a black, and the third 3 white and a black. It is not known in what order the bags are placed. A white counter is drawn from one of them, and a black from another. What is the chance of drawing a white counter from the remaining bag?

20. (43)

In the base of a given Triangle find a Point such that if from it two lines be drawn, terminated by the sides, one being perpendicular to the base and one to the left-hand side, they shall be equal.

[5/88

21. (21, 44)

Sum, (1) to terms, (2) to 100 terms, the series

[7/4/89

22. (21, 45)

Given the 3 'altitudes' of a Triangle: find its (1) sides, (2) angles, (3) area.

[4/6/89

23. (21, 46)

A bag contains 2 counters, each of which is known to be black or white. 2 white and a black are put in, and 2[Pg 6] white and a black drawn out. Then a white is put in, and a white drawn out. What is the chance that it now contains 2 white?

[25/9/87

24. (21, 47)

If, from the vertices of a triangle , the lines , , be drawn, intersecting at : find the ratio in terms of the two ratios , .

[5/86

25. (22, 48)

If '', '', '' represent proper fractions; and if, in a certain hospital, '' of the patients have lost an eye, '' an arm, and '' a leg: what is the least possible number who have lost all three?

[7/2/76

26. (48)

Within a given Triangle place a similar Triangle, whose area shall have to its area a given ratio less than unity, whose sides shall be parallel to its sides, and whose vertices shall be equidistant from its vertices.

[4/89

27. (22, 50)

There are 3 bags, each containing 6 counters; one contains 5 white and one black; another, 4 white and 2 black; the third, 3 white and 3 black. From two of the bags (it is[Pg 7] not known which) 2 counters are drawn, and prove to be black and white. What is the chance of drawing a white counter from the remaining bag?

[4/3/80

28. (22, 50)

If the sides of a given Triangle, taken cyclically, be divided in extreme and mean ratio; and if the Points be joined: find the ratio which the area of the Triangle, so formed, has to the area of the given Triangle.

[12/78

29. (51)

Prove that the sum of 2 different squares, multiplied by the sum of 2 different squares, gives the sum of 2 squares in 2 different ways.

[3/12/81

30. (52)

In a given Triangle, to place a line parallel to the base, such that if from its extremities lines be drawn, to the base, parallel to the sides, they shall be together double of the inscribed Line.

[15/3/89

31. (22, 53)

On July 1, at 8 a.m. by my watch, it was 8h. 4m. by my clock. I took the watch to Greenwich, and, when it said 'noon', the true time was 12h. 5m. That evening, when the watch said '6h.', the clock said '5h. 59m.'

On July 30, at 9 a.m. by my watch, it was 8h. 57m. by my clock. At Greenwich, when the watch said '12h. 10m.', the true time was 12h. 5m. That evening, when the watch said '7h.' the clock said '6h. 58m.'

My watch is only wound up for each journey, and goes[Pg 8] uniformly during any one day: the clock is always going, and goes uniformly.

How am I to know when it is true noon on July 31?

[14/3/89

32. (22, 53)

Sum the Series (1) to terms; (2) to 100 terms.

[7/4/89

33. (54)

Inscribe in a given Circle the maximum Tetragon having 2 parallel sides, one double the other.

34. (55)

From a given Point draw 2 Lines, one to the centre of a given Circle, and the other cutting off from it a Segment containing an angle equal to that between the Lines.

[21/12/74

35. (56)

With a given Triangle, to describe a Circle, cutting each side in two points, such that, if radii be drawn perpendicular to the sides, they are divided by the sides in given ratios.

[11/76

36. (57)

In a given Triangle, to draw a line, from a Point on one side of it, to a Point on the other side, perpendicular to one of these sides, and equal to the sum of the portions, of these sides, intercepted between it and the base.

[3/89

37. (22, 58)

Two given Circles intersect, so that their common chord subtends angles of 30° and 60° at their centres. What fraction of the smaller Circle is within the larger?

[12/91

[Pg 9]

38. (22, 60)

There are 3 bags, '', '', and ''. '' contains 3 red counters, '' 2 red and one white, '' one red and 2 white. Two bags are taken at random, and a counter drawn from each: both prove to be red. The counters are replaced, and the experiment is repeated with the same two bags: one proves to be red. What is the chance of the other being red?

[3/76

39. (22, 60)

and begin, at 6 a. m. on the same day, to walk along a road in the same direction, having a start of 14 miles, and each walking from 6 a. m. to 6 p. m. daily. walks 10 miles, at a uniform pace, the first day, 9 the second, 8 the third, and so on: walks 2 miles, at a uniform pace, the first day, 4 the second, 6 the third, and so on. When and where are they together?

[16/3/78

40. (61)

In a given Triangle, whose base-angles are acute, draw two lines, at right angles to the base, and together equal to the line drawn, from the vertex, at right angles to the base, and such that

(1) they are equidistant from the line drawn from the vertex;

(2) they are equidistant from the ends of the base.

[5/76

41. (23, 62)

My friend brings me a bag containing four counters, each of which is either black or white. He bids me draw two, both of which prove to be white. He then says[Pg 10] "I meant to tell you, before you began, that there was at least one white counter in the bag. However, you know it now, without my telling you. Draw again."

(1) What is now my chance of drawing white?

(2) What would it have been, if he had not spoken?

[9/87

42. (23, 63)

If the angles of a given Triangle be bisected, and if lines be drawn, through its vertices, at right angles to the bisectors, so as to form a fresh Triangle: find the ratio

of the area of this Triangle to the area of the given Triangle.

[17/5/78

43. (65)

From the ends of the base of a given Triangle draw two lines, intersecting, terminated by the sides, and forming an isosceles Triangle at the base, and a Tetragon, equal to it, at the vertex.

[2/82

44. (66)

If , be two numbers prime to each other, a value may be found for which will make a multiple of .

[18/3/81

45. (23, 67)

If an infinite number of rods be broken: find the chance that one at least is broken in the middle.

[5/84

46. (68)

In a given Triangle, whose base is divided at a given Point, inscribe a Triangle, having its angles equal to given angles, and having an assigned vertex at the given Point.

[19/11/87

[Pg 11]

47. (23, 69)

Solve the 2 Indeterminate Equations

and find the limits, if any, between which the real values lie.

[12/90

48. (70)

If semicircles be described, externally, on the sides of a given Triangle; and if their common tangents be drawn; and if their lengths be , , : prove that

is equal to the semiperimeter of the Triangle.

[9/2/81

49. (23, 72)

If four equilateral Triangles be made the sides of a square Pyramid: find the ratio which its volume has to that of a Tetrahedron made of the Triangles.

[16/11/86

50. (23, 72)

There are 2 bags, and , each containing 2 counters: and it is known that each counter is either black or white. A white counter is added to bag , the bag is shaken up, and one counter transferred (without looking at it) to bag , where the process is repeated, a counter being transferred to bag . What is now the chance of drawing a white counter from bag ?

51. (74)

[Pg 12]

From a given Point, in one side of a given Triangle, to draw a line, terminated by the other side, so that, if from its ends lines be drawn at right angles to the base, their sum shall be equal to the first line.

[12/81

52. (23, 75)

Five beggars sat down in a circle, and each piled up, in a heap before him, the pennies he had received that day: and the five heaps were equal.

Then spake the eldest and wisest of them, unfolding, as he spake, an empty sack.

"My friends, let me teach you a pretty little game! First, I name myself 'Number One,' my left-hand neighbour 'Number Two,' and so on to 'Number Five.' I then pour into this sack the whole of my earnings for the day, and hand it on to him who sits next but one on my left, that is, 'Number Three.' His part in the game is to take out of it, and give to his two neighbours, so many pennies as represent their names (that is, he must give four to 'Number Four' and two to 'Number Two'); he must then put into the sack half as much as it contained when he received it; and he must then hand it on just as I did, that is, he must hand it to him who sits next but one on his left—who will of course be 'Number Five.' He must proceed in the same way, and hand it on to 'Number Two,' from whom the sack will find its way to 'Number Four,' and so to me again. If any player cannot furnish, from his own heap, the whole of what he has to put into the sack, he is at liberty to draw upon any of the other heaps, except mine!"

[Pg 13]

The other beggars entered into the game with much enthusiasm: and in due time the sack returned to 'Number One,' who put into it the two pennies he had received during the game, and carefully tied up the mouth of it with a string. Then, remarking "it is a very pretty little game," he rose to his feet, and hastily quitted the spot. The other four beggars gazed at each other with rueful countenances. Not one of them had a penny left!

How much had each at first?

[16/2/89

53. (24, 76)

In a triangular billiard-table, a Point is given by its trilinear co-ordinates. A ball, starting from the given Point, strikes the three sides, and returns to the starting-point. Find, in terms of the trilinear co-ordinates and of the angles of the Triangle, the Point where the ball strikes the second side.

[6/4/89

54. (24, 78)

Cut off, from a given Triangle, by lines parallel to the sides, 3 Triangles, so that the remaining Hexagon may be equilateral. Also find the lengths of its sides in terms of the sides of the given Triangle: and the ratios in which the sides of the given Triangle are divided.

[18/4/86

55. (79)

Given three cylindrical towers on a Plane: find a Point, on the Plane, from which they shall look the same width.

[20/12/74

56. (24, 80)

Given the 3 altitudes of a Triangle: construct it.

[27/6/84

[Pg 14]

57. (25, 80)

In a given Triangle describe three Squares, whose bases shall lie along the sides of the Triangle, and whose upper edges shall form a Triangle;

(1) geometrically; (2) trigonometrically.

[27/1/91

58. (25, 83)

Three Points are taken at random on an infinite Plane. Find the chance of their being the vertices of an obtuse-angled Triangle.

[20/1/84

59. (25, 84)

Given a Tetrahedron, having every edge equal to the opposite edge, so that its facets are all (when looked at from the outside) identically equal: find its volume in terms of its edges.

[8/90

60. (25, 87)

Given a Triangle , and that its base is divided at in the ratio to : find the angles , .

[21/3/90

61. (89)

Prove that, if any 3 Numbers be taken, which cannot be arranged in A. P., and whose sum is a multiple of 3, the sum of their squares is also the sum of another set of 3 squares, the 2 sets having no common term.

[1/12/81

62. (91)

Given two Lines meeting at a Point, and given a Point lying within the angle contained by them: draw a line, through the given Point, and forming, with the given Lines, the least possible Triangle.

[12/76

[Pg 15]

63. (26, 92)

Given 2 equal Squares, in different horizontal planes, having their centres in the same vertical line, and so placed that the sides of each are parallel to the diagonals of the other, and at such a distance apart that, by joining neighbouring vertices, 8 equilateral Triangles are formed: find the volume of the solid thus enclosed.

[3, 4/9/90

64. (94)

Given a Triangle, and a Point within it such that its distance from one of the sides is less than its distance from either of the others: describe a Circle, with given Point as centre, such that its intercepts on the sides may be equal to the sides of a right-angled Triangle.

[18/12/74

65. (95)

How many shapes are there for Triangles which have all their angles aliquot parts of 360°?

[5/89

66. (26, 97)

Given that there are 2 counters in a bag, as to which all that was originally known was that each was either white or black. Also given that the experiment has been tried, a certain number of times, of drawing a counter, looking at it, and replacing it; that it has been white every time; and that, as a result, the chance of drawing white, next time, is . Also given that the same experiment is repeated times more, and that it still continues to be white every time. What would then be the chance of drawing white?

[9/89

[Pg 16]

67. (26, 100)

If a regular Tetrahedron be placed, with one vertex downwards, in a socket which exactly fits it, and be turned round its vertical axis, through an angle of 120°, raising it only so much as is necessary, until it again fits the socket: find the Locus of one of the revolving vertices.

[27/1/72

68. (26, 101)

Five friends agreed to form themselves into a Wine-Company (Limited). They contributed equal amounts of wine, which had been bought at the same price. They then elected one of themselves to act as Treasurer; and another of them undertook to act as Salesman, and to sell the wine at 10% over cost-price.

The first day the Salesman drank one bottle, sold some, and handed over the receipts to the Treasurer.

The second day he drank none, but pocketed the profits on one bottle sold, and handed over the rest of the receipts to the Treasurer.

That night the Treasurer visited the Cellars, and counted the remaining wine. "It will fetch just £11," he muttered to himself as he left the Cellars.

The third day the Salesman drank one bottle, pocketed the profits on another, and handed over the rest of the receipts to the Treasurer.

The wine was now all gone: the Company held a Meeting, and found to their chagrin that their profits (i. e. the Treasurer's receipts, less the original value of the wine) only cleared 6d. a bottle on the whole stock. These profits had accrued in 3 equal sums on the 3 days (i. e. the Treasurer's receipts for the day, less the original value of[Pg 17] the wine taken out during the day, had come to the same amount every time); but of course only the Salesman knew this.

(1) How much wine had they bought? (2) At what price?

[28/2/89

69. (26, 102)

If, from each of the angles of a given Triangle , taken cyclically, a certain proper fraction of it be cut off, the arithmetical values of the 3 fractions being represented by ', , '; and if it be given that the Triangle, formed by the lines so drawn, is similar to the given one, the angle, formed by the lines drawn from and , being equal to , and so on: find , , , as similar functions of a single variable. Also find the ratio which each side of the second Triangle bears to the corresponding side of the first.

[8/89

70. (27, 105)

Let an equilateral and equiangular Tetrahedron be placed with one facet in front: and suppose a series of triangles, equal to that facet, constructed in the Plane containing that facet, and having a base common with it; and that they are all wrapped round the Tetrahedron as far as they will go. Find (1) the locus of their vertices; (2) the situation of the vertex of the one whose left-hand base-angle is 15°; (3) the left-hand base-angle of the one which (wrapped round towards the right) covers portions of all four facets of the Tetrahedron, and whose vertex coincides with its vertex; (4) the left-hand base-angle of the one which (similarly treated) occupies all four facets, and then the front and right-hand facet for the second time, and whose vertex coincides with the distal vertex of the base of the Tetrahedron.

71. (108)

[Pg 18]

In a given Triangle place a Hexagon having its opposite sides equal and parallel, and three of them lying along the sides of the Triangle, and such that its diagonals intersect in a given Point.

[14/12/74

72. (27, 109)

A bag contains 2 counters, as to which nothing is known except that each is either black or white. Ascertain their colours without taking them out of the bag.

[8/9/87

FOOTNOTES:

[2] The numerals, placed in parentheses, indicate the pages where the corresponding matter may be found.

[Pg 19]

CHAPTER II.

Answers.

5. (2, 31)

Two-thirds.

6. (2,32)

Calling the sides '', '', '' and the lines '', '', '', we have

.

7. (2, 33)

Let , be given sides, and , the right s; and let , .

(1)

(2) .

[Pg 20]

8. (2, 34)

7 men; 2 shillings.

10. (3, 36)

Either 2 florins and a sixpence; or else a half-crown and 2 shillings.

11. (3, 36)

The required ratio is equal to

If , this =

12. (4, 37)

If = semi-perimeter, = area, = volume; then

13. (4, 38)

If '' = area of Tetragon whose vertices are the Centres and the Points of intersection; and if its sides be '', '', and its diagonal, joining the Centres, '': required area

16. (4, 40)

[Pg 21]

The first course gives chance = ; the second, . Hence the first is best.

18. (5, 41)

(1) Divide base , at , so that .

(2) At , , make right angles , ; and join cutting at , which is the required Point.

19. (5, 42)

Eleven-seventeenths.

21. (5, 44)

(1) ; .

22. (5, 45)

Calling the given altitudes ', , '; and the fraction

(1)

(2)

(3)

23. (5, 46)

Two-fifths.

24. (6, 47)

; whence any one can be found in terms of the other two.

[Pg 22]

25. (6, 48)

27. (6, 50)

Seventeen-twentyfifths.

28. (7, 50)

31. (7, 53)

When the clock says '12h. 2m. '

32. (8, 53)

37. (8, 58)

.

38. (9, 60)

Fortynine-seventytwoths.

39. (9, 60)

[Pg 23]

They meet at end of 2d. 6h., and at end of 4d.: and the distances are 23 miles, and 34 miles.

41. (9, 62)

(1) Seven-twelfths. (2) One-half.

42. (10, 63)

45. (10, 67)

47. (11, 69)

One set of values is 0, 0, 0.

A 2nd set is ; has any value.

A 3rd is ; has any value.

And the 4th set is , ; where has any value.

If has any positive value less than 4, and are unreal.

49. (11, 72)

Two.

50. (11, 72)

Seventeen-twentysevenths.

52. (12, 75)

2l. 18s. 0d.

[Pg 24]

53. (13, 76)

The portion, cut off from the second side, is equal to

54. (13,78)

Side must be divided at , , so that

and similarly for the other sides. Also each side of the Hexagon = .

56. (13, 80)

Draw , , equal to the given altitudes, so as to form right s at and : and produce , . Join , and draw to it. Join , and draw to it. With centre , and distance , describe a circle: with centre , and distance describe another: let them meet at : and join , . Triangle may be proved to be similar to required Triangle. The rest of the construction is obvious.

[Pg 25]

57. (14, 80)

(1) Geometrically.

If Squares be described externally on the sides of the given Triangle; and if their outer edges be produced to form a new Triangle; and if the sides of the given Triangle be divided similarly to those of the new Triangle: their central portions will be the bases of the required Squares.

(2) Trigonometrically.

If , , be the sides of the given Triangle, and its area; and if , , be the sides of the required Squares: then

58. (14, 83)

59. (14, 84)

Calling lengths of the 3 pairs of edges ', , ', and the corresponding s, in each facet, ', , '; volume =

60. (14, 87)

[Pg 26]

;

similarly, .

63. (15, 92)

If each side of each Square = 2, the volume =

66. (15, 97)

67. (16, 100)

If the centre of the horizontal facet be taken as the Origin, and if the -axis pass through one of the vertices of that facet, and the -axis be parallel to the opposite edge of that facet, and the -axis be perpendicular to that facet: and if the altitude (measured downwards) of the Tetrahedron be called '', and the intercept on the -axis be called '': the Equations to the Locus are

68. (16, 101)

(1) 5 dozen; (2) 8/4 a bottle.

69. (17, 102)

(1) ; ; .

(2) Calling new Triangle '',

[Pg 27]

70. (17, 105)

(1) Down the back-edge; up again; and so on. (2) about of the way down the back-edge. (3) About . (4) About .

72. (18, 109)

One is black, and the other white.

[Pg 28]

CHAPTER III.

Solutions.

1. (1)

Let , be the Nos.

Then .

Evidently '' is a form for the squares.

Also, if we write '' for '2' (which will not interfere with the problem, as we can divide by , and get), the above form becomes ''.

Now, as these are squares, their resemblance to

at once suggests itself; so that the problem depends on the known one of finding , , such that is a square; and we can then take as .

A general form for this is

Q. E. F.

[Pg 29]

2. (1)

(Analysis.)

Let be the Triangle, and the required line, so that .

From cut off equal to ; then .

Join , .

Now ;

Similarly ;

, are bisected by , , and is centre of escribed to .

Drop, from , s on , , ; then these s are equal.

Hence, if be joined, it bisects .

Hence construction.

(Synthesis.)

Bisect by : from draw , , , : also draw and equal to : and through draw , i. e. . Then shall be line required.

s at , , , are right, and ,

s , , are bisected by , .

Now ; it ; ;

Similarly ; .

Q. E. F.

[Pg 30]

3. (1)

Let be the Tetragon; and let the 3 sides, , , , be bisected by vertices of the Parallelogram .

Join .

, in Triangle , sides , are bisected at and ,

is parallel to ;

but is parallel to ;

is parallel to ;

Triangles , are similar;

now is half of ;

is half of .

Q. E. D.

4. (1)

Let be the given Triangle, and the required Triangle, so that , &c.

Evidently , are equally inclined to a line drawn,[Pg 31] from , ; and so of the others: i. e. these s bisect the s at , , ;

they meet in the same Point. Draw them; let them meet at ; and call the '', and so on.

Now ;

;

;

.

Similarly, .

Triangle is similar to Triangle ; and so of the others;

is foot of drawn, from , to . Hence the construction is obvious.

Q. E. F.

5. (2, 19)

At first sight, it would appear that, as the state of the bag, after the operation, is necessarily identical with its state[Pg 32] before it, the chance is just what it then was, viz. . This, however, is an error.

The chances, before the addition, that the bag contains (a) 1 white (b) 1 black, are (a) (b) . Hence the chances, after the addition, that it contains (a) 2 white (b) 1 white, 1 black, are the same, viz. (a) (b) . Now the probabilities, which these 2 states give to the observed event, of drawing a white counter, are (a) certainty (b) . Hence the chances, after drawing the white counter, that the bag, before drawing, contained (a) 2 white, (b) 1 white, 1 black, are proportional to (a) (b) ; i. e. (a) (b) ; i. e. (a) 2 (b) 1. Hence the chances are (a) (b) . Hence, after the removal of a white counter, the chances, that the bag now contains (a) 1 white (b) 1 black, are for (a) and for (b) .

Thus the chance, of now drawing a white counter, is .

Q. E. F.

6. (2, 19)

Call sides ', , ', and lines in question ', , '.

Now ;

[Pg 33]

To eliminate , , let us multiply by , , , so taken that

hence we may make ;

which gives lengths of sides.

Q. E. F.

7. (2, 19)

Let , be given sides, and , the right s; and let , .

Produce to meet -produced at .

similarly, ; which answers (1).

[Pg 34]

Q. E. F.

8. (2, 20)

Let = No. of men, = No. of shillings possessed by the last (i. e. the poorest) man. After one circuit, each is a shilling poorer, and the moving heap contains shillings. Hence, after circuits, each is shillings poorer, the last man now having nothing, and the moving heap contains shillings. Hence the thing ends when the last man is again called on to hand on the heap, which then contains shillings, the penultimate man now having nothing, and the first man having shillings.

It is evident that the first and last man are the only 2 neighbours whose possessions can be in the ratio '4 to 1'. Hence either

The first equation gives , i. e. , which evidently gives no integral values other than , .

[Pg 35]

The second gives , which evidently gives no positive integral values.

Hence the answer is '7 men; 2 shillings'.

9. (2)

Let , , be the given Lines, and the given Point; and join .

Through draw , , and bisected at ; from , , draw , , parallel to , and meeting , , at , ; join , and on it describe a semicircle cutting at ; and join , . Then is a right angle. From draw , , parallel to , .

Now Triangle has, to Triangle , the duplicate ratio of to ;

but so also has triangle to Triangle ;

also Triangles , , are equal, being on the same base , and having equal altitudes , ;

Triangles , are equal: and is evidently equal to ; it is a right angle.

Q. E. F.

[Pg 36]

10. (3, 20)

Call them , , ; and let .

The chance, that the pocket contains 2 balls, is ; and, if it does, the 'expectation' is the average value of

Also the chance, that it contains only one, is ; and, if it does, the 'expectation' is .

Hence total 'expectation' .

Hence the coins must be 2 florins and a sixpence; or else a half-crown and 2 shillings.

Q. E. F.

11. (3, 20)

Now

but

[Pg 37]

Q. E. F.

Cor. Let ; then .

12. (4, 20)

Let semi-perimeter, area, volume.

We know that ;

Q. E. F.

[Pg 38]

13. (4, 20)

Let , , be the centres of the Circles; , , their points of intersection; and the Tetragon whose area is required.

Let the sides of the Triangle be , , ; and its s , , .

Then , and .

Also ;

.

Hence area of Triangle ;

area of Tetragon = .

Now, writing '' for area of Triangle , we have

Q. E. F.

[Pg 39]

14. (4)

This simply expresses the identity

Q. E. D.

Numerical Examples (not thought out).

15. (4)

Let be an inscribed Tetragon. Join : and about Triangle describe a Circle.

Now, if this Circle does not pass through , let it cut , or produced, in or . Join , .

Then , or , is supplementary to ;

; which is absurd;

this Circle does pass through .

The same thing may be proved for any other Point on that portion, of the perimeter of the given Figure, which lies on the same side of as the Point .

Similarly for the other portion.

Hence the Figure is a Circle.

Q. E. D.

[Pg 40]

16. (4, 20)

The 'a priori' chances of possible states of first bag are ', ; , '. Hence chances, after putting in, are ', ; , '. The chances, which these give to the 'observed event', are 1, . Hence chances of possible states ', ', after the event, are proportional to 1, ; i. e. to 2, 1; i. e. their actual values are , .

Now, in first course, chance of drawing is ; i. e. .

And, in second course, chances of possible states ', ' are , : hence chance of drawing is ; i. e. .

Hence first course gives best chance.

Q. E. F.

17. (4)

(Analysis.)

Let be the given Triangle, and the line required.

From , draw , , parallel to the sides. Then .

Because is a Parallelogram, ;

similarly ;

.

Hence construction.

[Pg 41]

(Synthesis.)

Bisect s , , by , , meeting at ; through draw parallel to ; and from , , draw , , parallel to , .

Because is parallel to ,

Similarly .

.

Because , are Parallelograms,

Q. E. F.

18. (5, 21)

(1) Call required Point . From draw , sides. Join . From , , draw , . Call '', and ''.

Q. E. F.

[Pg 42]

(2) At , , make right angles , ; and join , cutting at . From draw , sides; and join .

, are they are ; ;

, are they are ; ;

;

is parallel to .

Q. E. F.

19. (5, 21)

Call the bags , , ; so that contains a white counter and a black one; &c.

The chances of the orders , , , , , , are, a priori, each. Since they are equal, we may, instead of multiplying each by the probability it gives to the observed event, simply assume those probabilities as being proportional to the chances after the observed event.

These probabilities are:—

[Pg 43]

Hence the chances are proportional to 4, 3, 8, 4, 9, 6; i. e. they are these Nos. divided by 34.

Hence the chance, of drawing a white counter from the remaining bag; is

20. (5)

(Analysis.)

Let be the given Triangle, and the required Point. Draw , and . Then .

Hence ;

, (draw ,)

Hence construction.

(Synthesis.)

From draw . Produce to , making equal to . Join . From draw parallel to ; and from draw , and .

Q. E. F.

[Pg 44]

21. (5, 21)

(1) The th term is ;

the th term is ;

and

Q. E. F.

(2) , to 100 terms,

now ;

;

and twice this ;

.

Q. E. F.

[Pg 45]

22. (5, 21)

Call given altitudes ', , '.

Now ;

;

(say);

.

Now ;

;

;

;

;

;

;

, substituting for , &c., and dividing by , [Pg 46];

.

Now , &c.; which answers (2).

Also ; and similarly ;

, \text{&c.}; which answers (1).

Also area ;

which answers (3).

Q.E.F.

23. (5, 21)

The original chances, as to states of bag, are

the chances, after adding 2 W and 1 B, are

Now the chances, which these give to the observed event, drawing 2 W and 1 B, are , , .

[Pg 47]

the chances, after this event, are proportional to , , ; i. e. to 2, 4, 1. Hence they are , , .

Hence the chances, as to states, now are

the chances, after adding , are

Now the chances, which these give to the observed event, of drawing , are 1, , .

the chances, after this event, are proportional to , , ;

i. e. to 6, 8, 1. Hence they are , , .

Hence the chance, that the bag now contains 2 white, is ; i. e. .

Q. E. F.

24. (6, 21)

Q. E. F.

[Pg 48]

25. (6, 22)

Let '' mean 'having lost an eye', '' 'having lost an arm', and '' 'having lost a leg'.

Then the state of things which gives the least possible number of those who, being and , are also , may evidently be found by arranging the patients in a row, so that the -class may begin from one end of the row, and the -class from the other end, and counting the portion where they overlap; and, the smaller the -class, the smaller will be this common portion: hence we must make the -class a minimum.

This may be done by re-arranging the patients, so that the -class may begin from one end of the row, and the -class from the other: and the least possible number for the -class is the common portion, i. e. , i. e. .

Then, as already shown, the least possible number for the -class is the common portion, i. e. , i. e. .

Q. E. F.

26. (6)

(Analysis.)

Let be the given Triangle, and the required one; and let the ratio, which has to , be ''; so that is less than 1.

Since , and that , , are parallel, it may easily be proved, by dropping perpendiculars from , , upon , which must necessarily be equal, that , , are equal.

[Pg 49]

Similarly, , , are equal; and so are , .

Call ''; then ;

;

.

Now , together ;

;

;

.

Hence, if , , be produced to meet at , Triangle will be isosceles, with a vertical equal to .

Now, if a Circle be drawn about , and its centre joined to and , the Triangle, so formed, will fulfil the same conditions;

hence the centre of this Circle will be ;

hence the construction.

(Synthesis.)

Bisect the sides, and draw perpendiculars, meeting at . Join to the vertices , . From cut off . From draw parallel to .

Then is easily proved equal to .

And if, from , , parallels to , , be drawn, it may easily be proved that they meet on , and that they are respectively equal to , .

Q. E. F.

[Pg 50]

27. (6, 22)

Call the bags , , .

If remaining bag be , chance of observed event = chance of drawing white from and black from + chance of drawing black from and white from :

i. e. it .

Similarly, if remaining bag be , it is ; and, if it be , it is .

chances of remaining bag being , , or , are as to to ; i. e. as 9 to 9 to 7. they are, in value, .

Now, if remaining bag be , chance of drawing white from it is ; chance, on this issue, is ; similarly, for , it is ; and, for , . And entire chance of drawing white from the remaining bag is the sum of these; i. e. .

28. (7, 22)

Let be the given Triangle; and let its sides be divided internally at , , , in extreme and mean ratio.

And let be the area of .

Let ; then ;

i. e. ;

[Pg 51]

, the other sign being excluded by the terms of the question.

Then area of Triangle

Similarly for and .

Hence the sum of these 3 Triangles , and area of Triangle .

Q. E. F.

29. (7)

This may be deduced from the identity

Now, if these last 2 sets are identical, must = ; for it cannot = ;

i. e., must = 0;

i. e., must = 0;

i. e., one or other of the first 2 sets is the sum of 2 identical squares.

Hence, contranominally, if each of the original sets consists of 2 different squares, their product gives the sum of 2 squares in 2 different ways.

Q. E. D.

[Pg 52]

30. (7)

(Analysis.)

Let be the Triangle: and suppose so placed that , , drawn parallel to the sides, shall together = .

By Euc. I. 34, , and :

Hence, if be cut off equal to half ,

Hence construction.

(Synthesis.)

In take any point : draw , , and equal to half : and join .

Similarly, in take any point : draw , , and equal to half : and join .

Produce , , to meet at : and through draw : and from , , draw , , the sides.

, by similar Triangles, ;

Similarly ;

; i. e. .

But, by Euc. I. 34, ;

.

Q. E. F.

[Pg 53]

31. (7, 22)

On July 1, watch gained on clock 5m. in 10h.; i. e. m. per hour; i. e. 2m. in 4h. Hence, when watch said 'noon', clock said '12h. 2m.'; i. e. clock was 3m. slow of true time, when true time was 12h. 5m.

On July 30, watch lost on clock 1m. in 10h.; i. e. 6sec. per hour; i. e. 19sec. in 3h. 10m. Hence, when watch said '12h. 10m.', clock said '12h. 7m. 19sec.'; i. e. clock was 2m. 19sec. fast of true time, when true time was 12h. 5m.

Hence clock gains, on true time, 5m. 19sec. in 29 days; i. e. 319sec. in 29 days; i. e. 11sec. per day; i. e. sec. in 5m.

Hence, while true time goes 5m., watch goes 5m. sec.

Now, when true time is 12h. 5m. on July 31, clock is (2m. 19sec. + 11sec.) fast of it; i. e. says '12h. m.' Hence, if true time be put 5m. back, clock must be put 5m. sec. back; i. e. must be put back to 12h. 2m. sec.

Hence, on July 31, when clock indicates this time, it is true noon.

Q. E. F.

32. (8, 22)

The th term is ;

Q. E. F.

[Pg 54]

Also .

Q. E. F.

33. (8)

Let ; .

Area .

let .

[Pg 55]

, omitting , and dividing by ,

(upper sign being inadmissible, though this was not thought out.)

Q. E. F.

34. (8)

(Analysis.)

Let be the given Point, and the centre of the given Circle. Join , and let be the required Line. From draw the Chord parallel to . Then . Hence Arc = Arc ; i. e. Arc is bisected by ; i. e. is on perpendicular from .

(Synthesis.)

Join . From draw perpendicular to . Join cutting Circle at . From draw parallel to .

It is easily proved that Arc = Arc . Hence Arc subtends, in the Circle, an angle = = .

Q. E. F.

[Pg 56]

35. (8)

Let be the given Triangle; and let the portions of the radii, outside the Triangle, have to the radius the given ratios , , . (N.B. , , , are supposed to be proper fractions.)

From draw , and ; and make have, to , the ratio . Through draw parallel to , and parallel to ; and join . From draw , and .

Then .

Similarly, draw so that the s, drawn from any Point of it to and , are in the ratio ; and produce to meet it at .

From draw , , , the sides.

Then .

Produce to , so that .

[Pg 57]

With centre , and distance , describe a Circle; and produce , , to meet it at , .

Similarly, .

But .

Similarly , and .

Q. E. F.

36. (8)

(Analysis.)

Let be required Line: and let at be right.

Cut off equal to : then .

Join , : then , and

From draw .

Then ; .

(Synthesis.)

Hence construction. Draw : bisect : at make . Let these lines meet at . Through draw .

Then ;

[Pg 58]

.

Q.E.F.

Limits of possibility:—

must not be ;

must not be ;

must not be half complement of ,

i. e. not .

37. (8, 22)

Let be the common chord, and , , the centres.

Let °, and °.

And let (which = .

And let .

[Pg 59]

Now ;

; ;

;

areas of Circles are and ;

areas of Sectors are and ;

their sum .

Again, area of Triangle ,

;

also area of Triangle ;

their sum .

Now the portion, of the smaller Circle, that is within the larger one, is the difference between these two sums;

it .

Hence its ratio, to the area of the smaller Circle, is this sum divided by ;

it ,

,

.

Q. E. F.

[Pg 60]

38. (9, 22)

Taking, in order, the bag from which this unknown counter is drawn, the bag from which a red one was twice drawn, and the remaining bag, we see that there are six possible arrangements of '', '', and '': viz.—

(1) , (4) ,

(2) , (5) ,

(3) , (6) .

Now the chance of the observed event is, in case (1), ; in case (2), ; in case (3), ; in case (4), ; in case (5), ; and in case (6), .

Hence the chances of existence, for these 6 states, are proportional to '12, 3, 18, 2, 9, 4'. Hence their actual values are ''.

Hence the chance of the unknown counter being red is the sum of ;

i. e. it is ; which .

Q. E. F.

39. (9, 22)

Let = no. of days.

Then ;

i. e. ;

; or .

Now the above solution has taken no account of the discontinuity of increase, or decrease of pace, and is the true solution[Pg 61] only on the supposition that the increase or decrease is continuous, and such as to coincide with the above data at the end of each day. Hence '4' is a correct answer; but '' only indicates that a meeting occurs during the third day. To find the hour of this, let no. of hours.

Now in 2 days has got to the end of 19 miles, to the end of , i. e. 20.

.

i. e. .

Hence they meet at end of 2d. 6h., and at end of 4d.: and the distances are 23 miles, and 34 miles.

Q. E. F.

40. (9)

(1) Let be the given Triangle, and the line from the vertex.

From draw , , parallel to the sides; and from and draw , , .

Then Triangles , , are similar to ;

.

Also, Triangles , , are equal and on the same base ,

their altitudes are equal; i. e. .

Q. E. F.

[Pg 62]

(2) Let be the given Triangle, and the line from the vertex.

Make ; from draw , , parallel to the sides; and from , , draw , , .

Then Triangles , , are similar to ;

.

.

Q. E. F.

41. (9, 23)

(1) As there was certainly at least one in the bag at first, the 'a priori' chances for the various states of the bag, ', , , ' were ', , , '.

These would have given, to the observed event, the chances '1, , , 0'.

Hence the chances, after the event, for the various states, are proportional to ', , '; i. e. to ', , '; i.e. to '2, 3, 1'. Hence their actual values are ', , '.

Hence the chance, of now drawing , is ''; i. e. it is .

Q. E. F.

[Pg 63]

(2) If he had not spoken, the 'a priori' chances for the states ', , , , ', would have been ''.

These would have given, to the observed event, the chances '1, , , 0, 0'.

Hence the chances, after the event, for the various states, are proportional to ', , '; i. e. to '1, 2, 1'. Hence their actual values are ', , '.

Hence the chance, of now drawing , is ''; i. e. it is .

Q. E. F.

42. (10, 23)

Let be the given Triangle. Bisect its angles, and draw s to them, forming the Triangle .

Now ; and so of the others.

[Pg 64]

Similarly, ;

Similarly, ;

area of ;

Q. E. F.

[Pg 65]

43. (10)

Let be the given Triangle; and let , , be the required lines, so that , and Tetragon = Triangle . And call the angle ''. It will suffice to calculate this angle.

Because Triangle Tetragon ,

.

Q. E. F.

[Pg 66]

44. (10)

Let be a No. not containing 2 or 5 as a factor, i. e. let it be prime to 10. Then, if be reduced to a circulating decimal, and that to a vulgar fraction, the digits of the denominator will be a certain number of 9's; i. e. it will be of the form .

And since this fraction , and that is prime to 10, and so prime to , the factor must be a multiple of .

This evidently holds good in any other scale of notation. Hence, if be the radix of the scale of notation, and a No. prime to , a value may be found for , which will make a multiple of .

Q. E. D.

EXAMPLES (not thought out).

(1) With radix 10, find a value, for , which will make a multiple of 7.

Ans. .

(2) Let the two given Nos. be 8, 9.

Taking 8 as radix, we get .

Ans. .

(3) Let the two given Nos. be 7, 13.

Taking 7 as radix, we get

Ans. .

[Pg 67]

45. (10, 23)

Divide each rod into parts, where is assumed to be odd, and the points of division are assumed to be the only points where the rod will break, and to be equally frangible.

Now, if ; then, when , ;

the chance that no rod is broke in the middle , when ;

i. e. it approaches the limit .

And Ans. .

Now . Hence if, in the series for , we call the sum of the odd terms '', and of the even terms ''; then ; and .

Q. E. F.

[N. B. What follows here was not thought out.]

[Pg 68]

46. (10)

Let be the given Triangle, and the given Point.

If we make a Triangle , having its angles equal to the given angles, and having as its assigned vertex, the Problem may be solved, if we can circumscribe, about the Triangle , a Triangle similar to .

Now we can construct, on , , , segments of Circles containing angles equal to , , . Hence the Problem may be solved, if we can place, in these Circles, a line , divided in the same proportion as .

[Pg 69]

This Lemma may be solved as follows. Let , , be the centres of the Circles. Join , and divide it, at , proportionally to .

Join ; through draw ; and from , , draw , , .

Now it may be easily proved that

But , , are doubles of , ;

Q. E. F.

[The construction is now obvious, viz. to join , , and produce them to meet, on the third Circle (as they may be easily proved to do), at ; then to divide , , at and , proportionally to , ; and then to join , .]

47. (11, 23)

By inspection, '' are one set of values.

Subtracting, we get ;

, unless , in which case .

Now, by (1), ;

, when , ;

, unless be infinite.

[Pg 70]

Similarly, by (2), , unless be infinite.

Hence, if be finite, and if , either or , and also either or ; i. e. either , or else . But the latter is excluded by our hypothesis. Hence . Hence ; i. e. either or , and the other may take any value.

This gives us 2 more sets of values, viz.

We have now to ascertain what happens when .

By (1), ;

.

Similarly, by (2), .

This gives us a 4th set of values, viz. ; where has any value.

Now and may evidently have any real values, but is restricted by the equation in which y cannot be real, unless . Hence may have any negative value, and any positive value that is not less than 4; but it cannot have any positive value, less than 4, without making unreal.

Q. E. F.

48. (11)

Let be the given Triangle, , , , the centres of the semicircles, and , , , the common tangents; so that , , and .

[Pg 71]

Join , ; and from draw . Hence .

Call sides of given Triangle ', , '.

Then , and ;

Q.E.D.

[Pg 72]

49. (11, 23)

Take, as unit, a side of one of the Triangles.

If the Tetrahedron be cut by a vertical Plane containing one of the slant edges, the section is a Triangle whose base is , and whose sides are ;

hence cosine of smaller base-angle

its sine its altitude;

and this is the altitude of the Tetrahedron;

volume of Tetrahedron .

Also altitude of Pyramid = altitude of Triangle whose base is , and whose sides are 1, 1;

i.e. it = ;

volume of Pyramid = .

Hence required ratio = .

Q. E. F.

50. (11,23)

At first, the chance that bag shall contain

[Pg 73]

, after adding a , the chance that it shall contain

hence the chance of drawing a W from it is the chance of drawing a is .

After transferring this (unseen) counter to bag , the chance that it shall contain

the chance of drawing a from it is

the chance of drawing a is .

Before transferring this to bag , the chance that bag shall contain

, after transferring it, the chance that bag shall contain

Hence the chance of drawing a is

i. e. the odds are 17 to 10 on its happening.

Q. E. F.

[Pg 74]

51. (12)

Let be the given Triangle, and the given Point.

(Analysis.)

Let be the line required. Draw , , the base. Then their sum is equal to .

Bisect at , and draw the base: then it is evident that is the of , , and is equal to half their sum; i. e. it is equal to half of . Hence a Circle, drawn with centre and at distance , will pass through and , and will touch the base at .

Through draw parallel to . Then is evidently bisected at . Also passes through the centre of the Circle. Hence, if be drawn (or), it is a chord of the Circle, and is bisected at . Produce to meet the base produced at . Hence cuts the Circle, and touches it at . But can be found, and can then be taken, so that sq. of may be equal to rect. of , .

(Synthesis.)

From draw , and produce it to meet the base produced at . Take , so that its square may be equal to rect. of , .

Bisect at , and from draw parallel to ; and from draw the base, to meet at . Join ,[Pg 75] and produce it to meet at , and draw , , the base.

Because , and that is parallel to ,

.

But .

Q. E. F.

[N. B. This proof is incomplete. I have assumed, without proving it, that . It may be proved thus. Because sq. of = rect. of , , is a chord of a Circle which touches the base at ; , which bisects it at right angles, passes through the centre. But also passes through the centre; is the centre; .]

52. (12, 23)

Let be the number of pennies each had at first.

No. (3) received , took out (2 + 4), and put in ; so that the sack then contained . Let us write '' for '.'

No. (5) received , took out (4 + 1), and put in enough to multiply, by , its contents when he received it. The sack now contained .

No. (2) took out (1 + 3), and handed on .

No. (4) took out (3 + 5), and handed on

No. (1) put in 2. The sack now contained .

Hence ;

Q. E. F.

[Pg 76]

53. (13,24)

Let be the given Triangle, and the given Point; and call its trilinear co-ordinates ', , '.

From draw , , , the sides, and therefore equal to , , . Produce and to and , making , and . From draw , and produce it to , making . Join , cutting in , and in . Join , cutting in . Join , .

The path of the ball is evidently ; and we have to calculate the length of .

Now .

First, to calculate .

[Pg 77]

From draw , , parallel to , ; from draw ; and from draw .

By similar Triangles, ;

;

.

Now ;

;

.

Similarly, ;

.

Again, ;

.

Similarly, ;

;

.

Now ,

;

,

.

Now ;

.

Q. E. F.

[Pg 78]

54. (13, 24)

It is evident that Triangle is similar to .

Let .

Now ; ;

;

;

;

but ;

;

;

(say).

Hence ·

;

.

Also ·

Q. E. F.

[Pg 79]

55. (13)

Let , , be the centres of the bases of the towers; and , , their radii. Suppose the required Point; and from draw a pair of tangents to each circle, and lines to the centres, which will evidently bisect the angles contained by the pairs of tangents.

Hence angles , , are equal;

;

i.e. ;

.

Draw a Line through , , and on it take Points , , such that .

Then the Semicircle, described on , is the locus of all Points whose distances, from and , are proportional to , .

Hence, if a Line be drawn through , , and a Semicircle described which shall be the locus of all Points whose distances, from and , are proportional to , ; the intersection of these two Semicircles will be the Point required.

Q. E. F.

[Note. "The locus of all Points whose distances &c.," if represented algebraically, is evidently a Circle, whose centre is on the Line through , , and which passes through and .]

[Pg 80]

56. (13, 24)

Draw , , , equal to the given altitudes, so as to form right s at and : and produce , . Join , and draw to it. Join , and draw to it. With centre , and distance , describe a circle: with centre , and distance , describe another: let them meet at : and join , .

Call the altitudes of , ', , '.

Now

= twice area of ;

also, taking as unit-line,

,

;

;

i.e. , , are proportional to given altitudes;

Triangle is similar to required Triangle.

The rest of the construction is obvious.

Q. E. F.

57. (14, 25)

(1) Geometrically.

Let be given Triangle.

(Analysis.)

Suppose the 3 Squares described, and that their upper edges form the Triangle . Join , , .

Now it is evident that, if be produced, the perpendiculars dropped, from any Point of it, upon , , will be proportional to , .

Similarly for and .

[Pg 81]

Hence these 3 Lines will meet at the Point from which the perpendiculars, dropped upon the sides of , are proportional to , , .

Hence, if Squares be described externally on the sides of , and if their outer edges be produced to form a new Triangle : this Triangle, with these 3 Squares, will form a Diagram wholly similar to that formed by the Triangle , with the 3 Squares inside it.

(Synthesis.)

Hence, if Squares be described externally on the sides of the given Triangle; and if their outer edges be produced to form a new Triangle; and if the sides of the given Triangle be divided similarly to those of the new Triangle: their central portions will be the bases of the required Squares.

Q. E. F.

(2) Trigonometrically.

Let , , be the sides of the given Triangle, and its area; and let , , be the sides of the required Squares.

It is evident that a Circle can be described about the Tetragon .

Hence .

Now, in Triangle , we know that

i. e. ;

.

Now is supplementary to ;

;

[Pg 82]

.

Now ;

.

Similarly, ·

But ;

;

;

i. e. .

Now it is evident that these Triangles are similar; so that

Hence, multiplying the last equation, throughout, by one or other of these equal fractions, we get

[Pg 83]

Hence, multiplying above and below by one or other of the equal fractions , , ,

Q. E. F.

58. (14, 25)

It may be assumed that the 3 Points form a Triangle, the chance of their lying in a straight Line being (practically) nil.

Take the longest side of the Triangle, and call it '': and, on that side of it, on which the Triangle lies, draw the semicircle . Also, with centres , , and distances , , draw the arcs , , intersecting at .

Then it is evident that the vertex of the Triangle cannot fall outside the Figure .

Also, if it fall inside the semicircle, the Triangle is obtuse-angled: if outside it, acute-angled. (The chance, of its falling on the semicircle, is practically nil.)

Hence required chance ·

[Pg 84]

Now let : then area of semicircle ; and area of Fig. ;

.

Q. E. F.

59. (14, 25)

and let s , , be equal to ', , '; and similarly for the s of the other facets.

From draw base-facet . Also draw , , , . And join , , .

It is easily proved that s , are right.

The required volume is . The area of is of course known. All we need is the length of . Now ; and evidently . Hence all we need is the length of .

[Pg 85]

Now this requires a preliminary Lemma, in itself a very pretty problem, viz.—

LEMMA (1).

Given, in Tetragon , sides , , and , and that s , are right: find .

Now ;

also ;

Q. E. F.

Hence this requires another Lemma, in order to find the value of (or , which will do as well, and makes a prettier problem).

[Pg 86]

LEMMA (2).

Given, in Triangle , sides , , and : find .

Q. E. F.

Hence, in Tetragon , we have by Lemma (1), and, by Lemma (2), ,

.

Now ;

,

;

therefore multiplied by ,

,

= ,

which is symmetrical, as it ought to be.

[Pg 87]

Now area of ;

hence volume of Tetrahedron

Q. E. F.

60. (14, 25)

Let , .

Now

;

[Pg 88]

Similarly, .

Q. E. F.

COROLLARIES.

(1) .

(2) .

(3) If Triangle be equilateral,

i. e., if be drawn to , ;

[Pg 89]

e. g., if , ·

(4) Let ;

then ,

;

;

from which, if were given, we could find from a Quadratic Equation.

I tried various values, to find one which would give rational values for and , and found that would do, as it leads to the Quadratic

in which becomes, after dividing all through by 6, , i. e. 49.

The Quadratic is ;

whence ; which solves the Problem 'Given a Triangle , having the tangents of its angles equal to 1, 2, 3: divide at , so that, if be joined, and drawn to it, the ratio may be '. The answer is 'Divide it so that '.

61. (14)

We know that the equation

'

'

is identically true.

[Pg 90]

Hence

;

, if () be a multiple of 3, so also is ;

is an integer;

and similarly for the other 2 fractions.

Also it may be proved that, if be equal to , or , or , then , , can be arranged in

First, let ;

then ; i. e. ;

secondly, let ;

then ; i. e. ;

thirdly, let ;

then ; i. e. .

And similarly for the other 2 fractions.

Hence, contranominally, if , , can not be arranged in , the 2 sets of squares have no common term.

Q. E. D.

[Pg 91]

Numerical Examples (not thought out).

62. (14)

Let , , be the given Lines, and the given Point.

Through draw parallel to ; from cut off equal to ; join , and produce it to meet at .

Because , and that is parallel to ,

.

Now let be any other line through ;

then .

Because, in Triangles , , , and

, and .

To each add Tetragon ;

.

And so of any other line through .

Hence is the least possible Triangle.

Q. E. F.

[Pg 92]

63. (15, 26)

Let each side of each Square = 2.

Then , ;

;

.

Take as origin, the -axis to , and the -axis to ; and let be part of the -axis.

Let equation to plane containing Triangle be

where is length of perpendicular dropped, from , upon this plane, and meeting it somewhere in .

Hence we can find from equation to , in the -plane, which will be

now this line contains , whose co-ordinates are [Pg 93] , and , whose co-ordinates are ;

,

and ;

;

;

;

.

Now area of ;

volume of pyramid, whose base is and whose vertex is ;

and there are eight such pyramids in the solid;

their sum = .

Also volume of pyramid, whose base is , and whose vertex is , = ;

and there are 2 such pyramids in the solid;

their sum = ;

volume of solid =

Q. E. F.

[Pg 94]

64. (15)

Let be the given Triangle, and the given Point; and let , its distance from , be less than either or , its distances from , .

Draw a line equal to , and it and equal to ; and join ; and about the Triangle describe a Circle; and place in it a line equal to ; and join .

Because sqs of , = sqs of , , and that is less than either or , is greater than either;

a Circle, with centre , and radius equal to , will cut all three Lines, in two Points each. Describe this Circle.

Hence , , , can be sides of a right-angled Triangle.

Q. E. F.

[Pg 95]

65. (15)

Calling the angles , , , of 360°, we must have

an Indeterminate Equation with 3 unknowns.

Evidently none of them can be so small as 2.

(1) Let ; then .

Now, if , will = :

hence can only be 1, or 2, or 3, or 6; and the same is true of .

(N.B. It is assumed that the fractions , are in their lowest terms.)

Let . Then .

Then its possible values are

,

,

, .

This gives 5 sets of values for , , , viz.:

(2) Let . Then , and, as before, can only be 1, or 2, or 4, and the same is true of . Hence the[Pg 96] possible values for are , so that

,

, .

This gives 3 more sets of values for , , viz.:

(3) Let ; then .

Hence denominator must contain factor "3", and can be only 1, or 2, or 5, or 10; and the same is true of .

Hence possible values of are , so that

,

, .

This gives 2 sets of values for , , viz.:—

but the latter (a fact overlooked in thinking out) we have had already.

(4) Let ; then .

Hence can be only 1, or 3, and the same is true of .

Hence possible values for are , so that

This gives 2 sets of values, viz.:—

but the latter (a fact overlooked in thinking out) we have had already.

[Pg 97]

There is no use in giving, to , any values greater than 6; for these would make ; so that one or other must be ; i. e. either or must , and we should get old values over again.

Hence there are 10 different shapes.

Q. E. F.

The 10 sets of angles (I am not certain that they were all thought out) are

(1) 120°, 30°, 30°;

(2) 120°, 40°, 20°;

(3) 120°, 45°, 15°;

(4) 120°, 36°, 24°;

(5) 120° °, °;

(6) 90°, 45°, 45°;

(7) 90°, 60°, 30°;

(8) 90°, 72°, 18°;

(9) 72°, 72°, 36°;

(10) 60°, 60°, 60°.

66. (15, 26)

Write for . Now the counters must be either both white, or one white and one black. Let chance of first condition be ; hence chance of second is . Hence chance of drawing white is .

Let a counter now be drawn and prove white; then chance of 'observed event,' in 1st condition, is 1, and, in 2nd condition, ;

[Pg 98]

Hence the chances, of the existence of these two conditions, are proportional to , ; i. e. are proportional to , ;

hence these chances actually are , ;

hence the chance of now drawing white,

Hence the effect of one repetition of the experiment has been to change into .

Hence a second repetition of it will change

We have now to discover the law (if there is one) for the series

regarding these as identical functions of 1, 2, 3.

We can write the 1st and 2nd term in the form of the 3rd, thus:—

and, by inspection, we see that each is of the form

where denotes the place of the term.

Suppose this law to hold for terms, what will be the effect of repeating the experiment once more?

[Pg 99]

We know that it changes into . Hence the new chance will be

i. e. the term of the series will follow the same law. But we know that the law holds for the , , and terms. Hence it holds universally.

Hence, after repetitions of the experiment, the chance of drawing white will be the term of the above series; i. e. it will be

Now, for , write .

Q. E. F.

Example—Let chance be ; and then let experiment be repeated 5 times more.

Here , ;

[Pg 100]

chance becomes , i.e. .

67. (16, 26)

Let be the socket. Revolve the Tetrahedron until the plane, in it, has taken the new position ; and let edge , in its new position, meet the socket-rim at . From draw -plane. Join , and produce it to . And draw , , the -ordinates of and .

Then co-ordinates of are , , .

Call , , ', '; and , , , ', , '; and ''.

It is evident that the vertical axis of the Tetrahedron always coincides with the -axis.

Hence moves on the surface of a cylinder,

i. e.

[Pg 101]

Now ;

;

Also Equation to is ;

;

Also, by similar s , ,

i. e. ;

;

but , and ;

;

i. e.

Equations (1) and (5) give the required Locus.

Q. E. F.

68. (16,26)

Let the Nos of bottles, taken out of the 3 days, be ', , '. Let each bottle have cost pence, and therefore be sold for pence.

Then the Treasurer's receipts, on the 3 days, were , , ; yielding, as profits (i. e. as remainders after deducting cost-price of bottles taken out), , , . Then these 3 quantities are equal.

[Pg 102]

Hence , and ;

total No. of bottles, being .

Now total profits are ; i. e. ;

profit, per bottle = ; and this must = 6;

.

Also ; i. e. ;

;

;

;

.

Now ;

;

No. of bottles = 60 or 48; but it is a multiple of 5;

it = 60.

Also ; i. e. ;

;

i. e. the wine was bought @ 8/4 a bottle, and sold @ 9/2 a bottle.

Q. E. F.

69. (17, 26)

§ 1. Let .

Then .

Now .

i.e.

Similarly,

[Pg 103]

and

From equations (1) and (3), and may be found in terms of : but these, taken along with , will not be similar functions of the single variable . We must have a certain function of , , , and (say); a similar function of , , , and ; and a similar function of , , , and ; i. e. we must have

Now we know, by (1), that ;

i. e. .

Now, as an experiment, let

then ;

These conditions will be fulfilled, if we make , and ; so that

which would make

Now this may evidently be simplified by omitting , which is constant; and we then have ; or, in a yet simpler form, by again subtracting , .

[Pg 104]

Q. E. F.

§ 2. We see that , so that is evidently equal to ; and so are s , .

This gives us a geometrical construction, viz. to draw lines from , , , so that each makes the same angle with the opposite side.

§ 3. Let us now ascertain the limits within which the value of must lie.

We know that .

Now ; i. e. ;

i. e. the supplement of ;

and of course this is true for each of the three angles , , ; i. e. if , , , be the order of the angles in a descending order of magnitude, supplement of .

Again .

Hence ; i. e. ;

and of course this is true for each angle.

Hence if , , , be the order in a descending order of magnitude, , and .

Q. E. F.

[Pg 105]

§ 4. We have now to ascertain the ratio which bears to .

In Triangle , whose s are , , , we have

, by symmetry, .

Now ;

Hence .

Q. E. F.

70. (17, 27)

Before folding the Plane containing the Triangles, the locus of their vertices is evidently a Line parallel to their common base. Hence, if the base of the Tetrahedron = 1, we may imagine a slip of paper, whose width is , attached to the front facet of the Tetrahedron, and wrapped round towards the right; and the upper edge of this slip will evidently be the[Pg 106] locus of the vertices. This slip may be conveniently regarded as divided into equilateral Triangles, placed base-downwards and base-upwards alternately, and it is evident that these Triangles will successively cover the facets of the Tetrahedron, in the order 'front, right side, base, left side,

front, &c.'; and its upper edge, made up of the bases of the inverted constituent Triangles, will evidently run as follows. Calling the successive Triangles, after the first (which occupies the front facet of the Tetrahedron), '' (base-up), '' (base-down), '' (base-up), '' (base-down), '' (base-up), and so on, the locus consists of the bases of , , &c. Now '' will occupy the right facet, its base coinciding with the back-edge of the Tetrahedron; '' will occupy the base of the Tetrahedron, its base coinciding with the front-edge; '' will occupy the left facet, its base coinciding with the back-edge; and so on. Hence the locus runs down the back-edge; up again; and so on. Which answers Question (1).

Q. E. F.

We may therefore, in answering the other three questions, consider the slip before it is folded, and calculate the positions of the vertices along its upper edge: and the problems thus become 'plane' ones.

[Pg 107]

(2) Gives us a right-angled Triangle, whose left-hand base-angle is 15°, and whose altitude is . We must calculate its base, and then, deducting half the base of the initial Triangle (i. e. deducting), we shall get the distance, measured along the upper edge of the slip, from the vertex of the initial Triangle to the vertex of the given Triangle; and from that we can calculate how many times we must go down and up the back-edge to reach it. Call the base of this right-angled Triangle '.' Then .

Now call 't'; then ;

Deducting , we get as the required distance.

Now &c.; &c.

Hence we must go down back-edge, up again, and then about down again. This answers question (2).

(3) We need to go down the back-edge, and up again; i. e. we must use up the upward bases of '' and ''. Hence the base of the required right-angled Triangle is . Hence the required left-hand base-angle is

Hence, for the required base-angle, we have ;

and the required angle is about of the way from one to the other. But the difference is almost exactly 6°. Hence we must add, to the lesser, about degrees, or . And the total will be about .

[Pg 108]

(4) Here the right-angled Triangle has, for its base, .

the required base-angle has, for its tangent,

,

= nearly . Now

Now &c°.; and we must go about half-way to the next angle, viz. &c°. The difference is about 6°; we must add about 3°. Hence the answer is about °.

71. (18)

Let be the given Triangle, and the given Point.

Bisect the sides of at , , ; and join these Points.

First, let be within the Triangle .

Draw parallel to , so that its distance from may be double the distance of from ; join , , and produce them to meet in , . From draw parallel to ; join , and produce it to meet at join .

[Pg 109]

Because is parallel to ,

;

,

.

Now the Triangles , are equal in all respects;

.

If lies on , and vanish, and the Hexagon becomes a Parallelogram.

If lies at , the Hexagon becomes the line .

If lies outside the Triangle , the Problem is insoluble.

Q. E. F.

72. (18, 27)

We know that, if a bag contained 3 counters, 2 being black and one white, the chance of drawing a black one would be ; and that any other state of things would not give this chance.

Now the chances, that the given bag contains , .

Add a black counter.

Then the chances, that it contains , , are, as before, , , .

Hence the chance, of now drawing a black one,

Hence the bag now contains (since any other state of things would not give this chance).

Hence, before the black counter was added, it contained , i. e. one black counter and one white.

Q. E. F.

THE END.

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TRANSCRIBER’S NOTES

For the sake of clarity, the notation used by the author for and has been replaced with modern terminology.

Due to a lack of distinction between 'a' and '' in the printed version, certain expressions have been corrected to avoid potential mathematical errors.